

Technology Stack

Date	16 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	HTML, CSS, Bootstrap	Used to create a simple, responsive, and user-friendly interface for entering HDI parameters and displaying prediction results.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python, Flask	Flask handles user requests, processes input data, and integrates the Machine Learning model to generate HDI predictions efficiently.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	CSV Dataset (Pandas)	The HDI dataset is stored in CSV format and managed using Pandas for data preprocessing, analysis, and model training.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Localhost / Flask Development Server (or Render/Heroku if deployed)	Used to run and test the application during development and demonstrate the project in a web environment.
5	Version Control & CI/CD	Git & GitHub	Enables version control, code management, collaboration among team members, and maintains project history.

6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, Anaconda, Jupyter Notebook	Used for machine learning model development, data preprocessing, visualization, and model evaluation.
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